

PML Absorbing Boundaries

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Abstract It has been previously demonstrated that no reflection is generated when elastic (or electromagnetic) waves enter a region with Perfectly Matching Layer (PML) absorbing conditions in a continuous medium. The practical application of PMLs, however, is in numerical modeling, where the medium is discretized by either a finite-element or a finite-difference scheme thus introducing a reduced amount of reflection. In such a case what is the practical and quantitative efficiency of PML absorbing boundaries? Assuming a regular spatial mesh, we start by evaluating analytically the reflection of body waves introduced by the discrete transition toward PML properties, under variable angle of incidence and wavelength. We then extend our evaluation with numerical tests for both body and Rayleigh waves. Surprisingly enough, the absorption remains equally efficient at wavelengths far larger than the PML thickness itself. As a consequence, the PML thickness can be kept minimal even for studies involving relatively low frequencies, and no rescaling with model size is required. Another pleasant feature is that it is all the more efficient at shallow angles of incidence. Finally, we show through numerical examples that a major advantage of using PMLs is their efficiency in absorbing Rayleigh waves at the free surface, a point where more classical methods perform rather poorly. Although previous authors essentially limited the description of their discrete implementation to 2D, we develop to some level of detail a 3D finite-difference scheme for PMLs and provide numerical examples.

Introduction

The Perfect Matching Layer (PML) absorbing conditions were first developed for numerical simulations of electromagnetic wave propagation to avoid reflections at the boundaries of the discrete model (Berenger, 1996). Subsequently, the method was successfully extended to the field of elasticity, where a very similar system of equations governs the dynamics (Chew and Liu, 1996; Collino and Tsogka, 2001), poroelasticity (Zeng *et al.*, 2001) and anisotropic media (Becache *et al.*, 2001). The evolution of this tool, first born in the electromagnetic field and then adopted by developers of elastodynamics modeling, is reminiscent of what happened before for staggered-grid, finite-difference schemes (Yee, 1966; Madariaga, 1976; Virieux, 1986).

As shown by Chew and Liu (1996), the implementation of PML is equivalent to the introduction of a system with stretched coordinates, where the stretching coefficient along a prescribed axis is a complex number with an evanescent imaginary part, whereas no stretching is applied along other axes. This causes a selective attenuation of the fields propagating in the prescribed direction. It is, in practice, an anisotropic sponge that turns out to be very efficient for wave absorption at the boundaries of numerical models.

Less recent methods, essentially based on a paraxial approximation as described by Clayton and Engquist (1977) and later improved by Higdon (1991), are still widely used

for wave absorption at the boundaries of numerical models. Although they are attractive for their simplicity of implementation, the minimal number of grid nodes required, and their satisfactory absorption of body waves under a wide range of incidence angles, they handle surface waves, which are essential and frequently encountered in geophysical applications, poorly.

On the other hand, PMLs are slightly more complex to implement and they require a finite number of nodes (usually 5 to 10), but they provide an excellent absorption of body waves, do not lose efficiency at shallow angles, and mostly, they are very effective even with surface waves. The spurious reflections at model boundaries can be made arbitrarily small by increasing the thickness of the PML layer at the cost of additional computation. However, an absorption level satisfactory for most practical uses can be obtained at a negligible cost, with respect to the computation in the bulk of the medium. As the medium becomes larger, the thickness of PML layers needs not increase, so that the cost of PML becomes rapidly negligible.

We further note that recent theoretical insights about exact local boundary conditions for the simulation of an infinite medium have been proposed by Grote and Keller (2000). Their study essentially involves an analysis in spherical harmonics at the boundary of a spherical model, with

no additional nodes outside. The concept is interesting indeed, but of limited practical applicability in the field of Cartesian finite-difference grids in its present form.

In the present study we present some features of PMLs that had not been investigated in previous literature, although they have implications of great practical interest. We show that PMLs are efficient even in the presence of a free surface with propagation of strong Rayleigh waves, although they were theoretically developed within the framework of body waves (within an infinite homogeneous model). This point is essential because it constitutes a considerable advantage of PMLs with respect to other absorbing boundary conditions. We extend our analysis of inhomogeneous waves to the case of an internal contrast in the medium properties and show that the absorbing boundary conditions remain efficient. Further, we find that the large wavelengths, even those much larger than the PML thickness, are effectively absorbed; this important feature was apparently unnoticed in previous tests. In addition, the dispersion analysis of our study is specific to the finite-difference grid, whereas previous PML dispersion studies were developed for finite-element formulation. Finally, we describe the implementation for a fully 3D medium by extending the scheme previously proposed in 2D by Collino and Tsogka (2001).

In the coming sections, we will develop the following arguments. We briefly recall the elastodynamics equations for stretched coordinates and the splitting of all the field variables into three components, resulting in three orthogonal gradients, according to Chew and Liu (1996). We develop the PML equations for a 3D, finite-difference scheme and briefly comment on interfacing with the bulk medium. As far as body waves are concerned, we extend classical dispersion–stability analysis to the special equations governing the PML. We then proceed to an analytical estimation of the spurious reflection generated at the discrete transition between the medium and the PML, assuming a step transition on the discrete mesh (i.e., a transition to a PML with constant properties) and derive theoretical wavelength and angular dependence. We then evaluate the reflection for more complex cases (a PML with a gradient) with the help of numerical tests and also provide angular and wavelength dependence. Because we are unaware of any analytical method for computing the reflection of inhomogeneous waves at lateral contrasts, we provide numerical estimates for the case of Rayleigh waves and investigate its wavelength dependence. We compare the results obtained for Rayleigh waves absorption with PML to those obtained with a paraxial approximation. Then we present the case of a half-space with a contrast in properties across a horizontal interface.

PML and Its Implementation in Finite-Difference Algorithm

The aim of a perfectly matched layer is to create an absorbing layer of finite thickness at the interface with the

model, introducing the minimum possible spurious reflection. This boundary condition can be obtained through the stretching of the coordinates by a complex factor S inside the PML. Consider the elastodynamics equation

$$\rho \partial_{tt} \mathbf{u} = \nabla \cdot \boldsymbol{\sigma} \quad (1)$$

where ρ is the mass density, $\mathbf{u}$ is the displacement field, and $\boldsymbol{\sigma}$ is the stress tensor. The external forces don't take part in the formulation of the problem and can be neglected. A similar equation can be used in the PML, if the derivative operator is substituted with the form:

$$\nabla_e = \frac{1}{S_x} \mathbf{i} \partial_x + \frac{1}{S_y} \mathbf{j} \partial_y + \frac{1}{S_z} \mathbf{k} \partial_z \quad (2)$$

where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the axes unit vectors and the S_i values are the stretching functions.

In a real finite-difference model the interfaces are orthogonal to one of the unit vectors of the chosen reference frame. If the interface between the medium and the PML is orthogonal to the x axis, for example, then $S_y = S_z = 1$ and S_x can be chosen conveniently for progressively reducing the body waves inside a layer of finite thickness without creating reflections at the interface with the medium (Chew and Liu, 1996). If we consider a complex wavenumber, the same conditions apply to inhomogeneous waves. To exponentially attenuate a plane wave, S_x should assume a complex value. In the frequency domain we choose:

$$S_x = 1 - \frac{i\alpha^x(x)}{\omega} \quad (3)$$

where $\alpha^x(x)$ is a real, scalar function of the distance x from the interface and ω is the pulsation. For continuity reasons, $\alpha^x(x)$ should be zero on the interface between medium and PML. Moreover, in our example the boundary is perpendicular to x so that $\alpha^y = \alpha^z = 0$. Reflections caused by the numerical scheme, in practice, are lowest for smooth $\alpha(x)$ functions, many times differentiable at the origin. A term in ω^{-1} in the Fourier domain reflects a constant in the time domain. In this manner the waves are attenuated, inside the PML, in proportion to $e^{-k/\omega}$ (Collino and Tsogka, 2001).

Let us now separate velocity $\mathbf{v}$ into three partitions $\mathbf{v}^{k=1,2,3}$, where the superscript k indicates that the single contribution of space derivative in the direction k has been isolated. From equations (1)–(3) we can write, in the Fourier domain, each velocity component v_i as:

$$v_i = \sum_k v_i^k, \quad (4)$$

where $\rho[i\omega + \alpha^k(x)]v_i^k = \sigma_{ik,k}$. Clearly, $i\omega$ indicates the time derivative. (Note that no implicit summation over repeated indexes is assumed in this expression.) Again, we

note that α^k is nonzero only when k corresponds to the normal of the medium boundary. Analogous relationships can be obtained for the constitutive law, by defining the stress partitions as:

$$[i\omega + \alpha^k(x)]\sigma_{ij}^k = \lambda\delta_{ij}\delta_{kl}u_{l,l} + \mu(\delta_{kj}u_{i,i} + \delta_{ki}u_{j,j}) \quad (5)$$

$$\sigma_{ij} = \sum_k \sigma_{ij}^k$$

where λ and μ are Lamé constants. Inside the model, of course, momentum and constitutive law reduce to the standard expressions:

$$\rho v_i = \sigma_{ij,j} \quad (6)$$

$$\sigma_{ij} = \lambda \delta_{ij} u_{l,l} + \mu(u_{i,j} + u_{j,i}). \quad (7)$$

(This time summation over repeated indexes is assumed.)

It appears from equation (4) that this model would require knowledge of stress, velocity, and acceleration at the same instant, which is not the case in the finite-difference staggered grid. However, a closer analysis shows that velocity can be written as a second-order interpolation of the values at the previous and next time step, the same values classically used in the second-order extrapolation expression. Thus, the classical scheme can be extended using the same nearest-neighbor values without loss of precision.

With this in mind the finite-difference scheme for the velocity-stress, staggered-grid formulation of the elastodynamics equation can be extended to the following PML formulation:

$$\rho(D^t + \alpha^k M^t)v_i^k = D_l \delta_{kl} \sigma_{il}$$

$$v_i = \sum_k v_i^k,$$

through the use of discrete operators D^t and D_l defined lower. We expand the expressions for the velocity component v_z :

$$\rho(D^t + \alpha^x M^t)v_z^x = D_x \sigma_{xz}$$

$$\rho(D^t + \alpha^y M^t)v_z^y = D_y \sigma_{yz}$$

$$\rho(D^t + \alpha^z M^t)v_z^z = D_z \sigma_{zz}$$

$$v_z = v_z^x + v_z^y + v_z^z.$$

The equivalent expressions for v_x and v_y are straightforward. Similarly, the discretization of Hooke's law can be split into:

$$(D^t + \alpha^k M^t)\sigma_{ll}^k = \lambda D_k v_k, \quad k \neq l$$

$$(D^t + \alpha^k M^t)\sigma_{ll}^k = (\lambda + 2\mu) D_k v_k, \quad k = l$$

$$\sigma_{ll} = \sum_k \sigma_{ll}^k,$$

for the diagonal stress components and into:

$$(D^t + \alpha^k M^t)\sigma_{ik}^k = \mu D_k v_i, \quad k \neq i$$

$$\sigma_{ij} = \sigma_{ij}^i + \sigma_{ij}^j, \quad i \neq j$$

for the shear components. For clarity, we expand the expressions for the diagonal component σ_{zz} :

$$(D^t + \alpha^x M^t)\sigma_{zz}^x = \lambda D_x v_x$$

$$(D^t + \alpha^y M^t)\sigma_{zz}^y = \lambda D_y v_y$$

$$(D^t + \alpha^z M^t)\sigma_{zz}^z = (\lambda + 2\mu) D_z v_z$$

$$\sigma_{zz} = \sigma_{zz}^x + \sigma_{zz}^y + \sigma_{zz}^z$$

and for the shear component σ_{yz} :

$$(D^t + \alpha^y M^t)\sigma_{yz}^y = \mu D_y v_z$$

$$(D^t + \alpha^z M^t)\sigma_{yz}^z = \mu D_z v_y$$

$$\sigma_{yz} = \sigma_{yz}^y + \sigma_{yz}^z$$

and the remaining stress components are readily derived along the same lines.

Assume in the following that p , q , and r represent the spatial grid indexes along x , y , and z , respectively, and n represents the time iteration. Then the discrete operators, applied to any scalar field variable ϕ , are defined as:

$$D^t \phi = (\phi_{p,q,r}^{n+1/2} - \phi_{p,q,r}^{n-1/2})/\Delta t$$

$$D_x \phi = (\phi_{p+1/2,q,r}^n - \phi_{p-1/2,q,r}^n)/\Delta x$$

$$D_y \phi = (\phi_{p,q+1/2,r}^n - \phi_{p,q-1/2,r}^n)/\Delta y$$

$$D_z \phi = (\phi_{p,q,r+1/2}^n - \phi_{p,q,r-1/2}^n)/\Delta z$$

and the arithmetic mean operator, times the attenuating scalar function, yields:

$$\alpha^k M^t \phi_{p,q,r}^n = \alpha_{p,q,r}^k (\phi_{p,q,r}^{n+1/2} + \phi_{p,q,r}^{n-1/2})/2$$

Note that the spatial derivative operators defined previously, for use inside the PML, are of the second order. However, we experienced no problem when coupling second-order PMLs with fourth-order operators inside the model. Tests with fourth-order operators in both PML and medium, however, resulted in instability and we dropped them, noting that fourth order in the PML was superfluous because the signal is rapidly being attenuated and discarded anyway. Again, if direction k is perpendicular to the medium boundary, only the coefficient α^k is nonzero, and, in practice, $\alpha_{p,q,r}^k$ can be reduced to α_s^k if s is the spatial index corresponding to k , because α remains unchanged along other directions. Let's illustrate this by taking, again, a PML orthogonal to the x

axis; in that case, α reduces to a function of only p (e.g., x) with a single nonzero component, α_p^x .

In our 3D finite-difference model we do not construct PMLs with curved boundaries, but only define interfaces orthogonal to the coordinate axes. Inside PMLs, only waves propagating in the outgoing direction are attenuated, whereas those propagating parallel to the interfaces are not affected. The latter may bounce back and forth and increase indefinitely as new waves are merging into the PML from the model. To avoid such problem, a 2D attenuation is defined in the corners where PMLs from perpendicular boundaries overlap, where two of the α^k components are nonzero. Finally, the cubic model is completely bounded by PMLs.

The last theoretical question is related to the definition of the function $\alpha(x)$ (e.g., α_p , where we omit the superscript for convenience). Because it must have inverse time dimensions and depend on the distance from the interface, we propose the ansatz:

$$\alpha_q = A \frac{c_p}{h\Delta x} \left(\frac{q}{h}\right)^n \quad (8)$$

where q is the distance in grid points from the interface, c_p is the compressional wave velocity, h is the total PML thickness in grid points, and A is a constant, dimensionless factor.

From numerical tests, a value of about $A = 5$ yields the most satisfactory results, whereas any of the values $n = 2, 4, 5$ seem to work well, as opposed to the 2D case where $n = 5$ is much better (see Numerical Tests for Body Waves). (Collino and Tsogka [2001] suggest a value $n = 4$ for the 2D scheme, but it turns out to be less efficient in 3D.) The value of A can be related to the minimum value of signal attenuation of the signal in the layer, which is of the order of 10^{-3} .

Stability Analysis

We shall now derive the dispersion relationship under PML conditions. Recalling the definition that a superscript indicates a particular partition of a quantity (i.e., $\mathbf{v}^x$ is the segregation of the field $\mathbf{v}$ deriving from the single gradient component ∇_x), we can express stress and velocity partitions as plane waves in the following manner:

$$\mathbf{v}^x = \mathbf{V}^x e^{i(\omega t - \mathbf{k} \cdot \mathbf{x})} \quad (9)$$

$$\boldsymbol{\sigma}^x = \boldsymbol{\Sigma}^x e^{i(\omega t - \mathbf{k} \cdot \mathbf{x})} \quad (10)$$

where $\mathbf{V}$ and $\boldsymbol{\Sigma}$ represent the maximum values of the fields, ω the pulsation (related to the frequency), and $\mathbf{k}$ the wave-number. To obtain the dispersion relationship for the velocity–stress formulation in the PML, equations (4) and (5) must be solved simultaneously. If we substitute the plane wave solution into equation (4) for the component v_x^x , we obtain:

$$\rho \left[2i \frac{\sin(\omega\Delta t/2)}{\Delta t} + \alpha(x) \cos(\omega\Delta t/2) \right] V_x^x = -2i \frac{\sin(k_x\Delta x/2)}{\Delta x} \Sigma_{xx}$$

or equivalently

$$\rho 2i \frac{\sin(\omega\Delta t/2)}{\Delta t} (1 - i R_x) V_x^x = -2i \frac{\sin(k_x\Delta x/2)}{\Delta x} \Sigma_{xx}$$

where

$$R_x = \frac{\Delta t \alpha(x)}{2} \cot(\omega\Delta t/2)$$

Let us now pose:

$$\begin{aligned} \Omega &= \omega \operatorname{sinc}(\omega\Delta t/2) \\ K_x &= k_x \operatorname{sinc}(k_x\Delta x/2) \\ \hat{K}_x &= \frac{K_x}{1 - iR_x}, \end{aligned} \quad (11)$$

then, the discrete derivative becomes formally equivalent to the classical derivative and the partition is intrinsic in the definition of $\hat{K}_i$. The resulting expression for V_x is

$$\rho i \Omega V_x = -(i \hat{K}_x \Sigma_{xx} + i \hat{K}_y \Sigma_{xy} + i \hat{K}_z \Sigma_{xz})$$

Analogous relations can be obtained for other components of the velocity and for all the stress components. Because the dispersion relation in the discrete case is unchanged with respect to the analytical case, we can proceed in the same manner, extracting the values of $\boldsymbol{\Sigma}$ from the constitutive law, replacing them in the momentum law, and solving the linear system for the unknowns V_i . To obtain a nontrivial solution, the rank of the incomplete matrix cannot be maximum. The solution of this equation in the unknown Ω , as a function of $\hat{\mathbf{K}}$ is the standard dispersion relationship (Aki and Richards, 1980):

$$\Omega^2 = c_p^2(\hat{K}_x^2 + \hat{K}_y^2 + \hat{K}_z^2) = c_p^2(\hat{\mathbf{K}} \cdot \hat{\mathbf{K}})$$

for the P wave and, for the S wave (both SV and SH):

$$\Omega^2 = c_s^2(\hat{K}_x^2 + \hat{K}_y^2 + \hat{K}_z^2) = c_s^2(\hat{\mathbf{K}} \cdot \hat{\mathbf{K}})$$

In the standard finite-difference scheme we impose that the solutions for ω are real, for any value of the wavenumber $\mathbf{k}$. In PMLs, it is impossible to solve the equation analytically, because the values of $\hat{K}_i$ themselves intrinsically contain Ω . If the zeros of ω lie in the negative half-plane the method becomes unstable. Analyzing the case in which the PML is joined along the yz plane, the relationship for the P wave is

$$\Omega^2 = c_p^2(\hat{K}_x^2 + K_y^2 + K_z^2)$$

or, in extension:

$$\begin{aligned} & \sin(\omega\Delta t/2) \\ &= \frac{c_p}{\Delta x} \sqrt{\frac{\sin^2(k_x\Delta x/2)}{(1 - iR_x)^2} + \sin^2(k_y\Delta x/2) + \sin^2(k_z\Delta x/2)}. \end{aligned}$$

From this function, Appendix A shows that stability is achieved for the same interval of P as in the classical finite-difference scheme.

It can be observed that the scheme is stable for a Courant number of the order of $P = 0.3$, and for values of $\mathbf{k}$ ensuring a minimum of 8 grid points per wavelength. The same value of P also warrants stability and weak numerical dispersion in standard finite-difference schemes.

The Courant number relating time and space sampling to the maximum value of wave velocity in the medium can be defined in 3D (Mozco *et al.*, 2000) as:

$$P = \frac{\Delta t V_p}{\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}} = \frac{\Delta t V_p}{\Delta x \sqrt{3}}.$$

In theory, it is required that $P < 1$ to ensure stability of a second-order algorithm at any frequency. In most practical cases P must be several times smaller.

In the case of the anisotropic continuous medium, it has been shown that the stability condition may not be met (Becache *et al.*, 2001) inside the PML.

Is PML a Nonreflecting Interface?

The perfectly matched layers are analytically constructed to attenuate the signal exponentially without introducing reflections at the interface with the effective medium. This condition is verified for any frequency, angle of incidence, and type of wave, provided that an appropriate choice of stretching coefficients is made. However, in the numerical scheme this condition is not verified because the discrete approximation introduces finite reflections, if only because of numerical dispersion and the finite contrasts in properties between one grid node and the next. We shall now analyze and predict the amplitude of the reflection due to the finite contrast between the model and a homogeneous PML on a discrete spatial mesh. This, in fact, corresponds to the study of a single-step transition in the properties of the discretized medium. Remarking that the actually implemented PMLs usually have variable properties, one may question the significance of this analysis, but again, a discretized gradient in properties can be considered as a series of steps. Because we are dealing with a linear problem, we can apply the principle of superposition, and conclude that the nature of PML behavior remains unchanged for either single or multiple steps. However, we note that some subtle effects may take place, owing to second-order interaction between the various steps

and phase delays between the effect of each single step, all of which shall be accounted for in the numerical estimations to be found in the next section.

Let us start with the simplest case of an SH plane wave propagating in the xz plane, because no conversion into other types of waves is possible. Let the PML boundary lie orthogonal to the x direction and position the origin of the reference frame at the PML-model interface, with no loss of generality. The particle velocity is only along the y direction and can be expressed as:

$$v_y = e^{i(\omega t - k_x^s x - k_z^s z)} + R e^{i(\omega t + k_x^s x - k_z^s z)}$$

where R is the reflection coefficient and $\mathbf{k}^s$ is the wavenumber of the S -incidence wave. Analogously, the expression for the velocity field inside PML is:

$$v_y' = T e^{i(\omega t - k_x'^s x - k_z'^s z)}$$

where T is the transmission coefficient and the prime distinguishes the variables from their equivalent values outside the PML. We take a unit amplitude for the incident wave. In a continuous medium, the reflection/transmission coefficients can be obtained by imposing the continuity condition of velocity and stress across the interface. Owing to the finite-difference grid staggering, we cannot impose both conditions, because some of the variables are not defined at the interface. With our choice of reference frame, only v_y is located at the interface. However, we remark that a condition can be imposed on the derivative of σ_{xy} . (A similar procedure is classically used to implement the free-surface condition, where grid staggering precludes the definition of all the traction components at the very surface.)

Continuity of v_y yields to:

$$1 + R = T \quad (12)$$

The shear stress σ_{xy} can be computed using the constitutive law separately in the two layers:

$$\begin{aligned} \sigma_{xy} = & -\frac{K_x}{\Omega} \mu e^{i(\omega t - k_x^s x - k_z^s z)} \\ & + \frac{K_x}{\Omega} \mu R e^{i(\omega t + k_x^s x - k_z^s z)} \quad (13) \end{aligned}$$

$$\sigma_{xy}' = -\frac{\hat{K}_x'}{\Omega} \mu T e^{i(\omega t - k_x'^s x - k_z'^s z)} \quad (14)$$

where μ , the shear modulus, is the same in the both layers. Differentiation of the constitutive law with respect to x yields

$$\partial_{xt}^2 \sigma_{xy} = \mu \partial_{xx}^2 v_y,$$

which, in the sense of the finite differences, corresponds to

$$\frac{1}{\Delta x} [(\sigma'_{xy})_{x=\Delta x/2} - (\sigma_{xy})_{x=-\Delta x/2}] = -\hat{K}'_x \mu (v'_y)_{x=0}. \quad (15)$$

Substitution of (13) and (14) into (15) yields the second condition for the transmission/reflection:

$$\frac{1}{\Delta x} (\hat{K}'_x T e^{-ik'_x \Delta x/2} - K_x e^{ik_x \Delta x/2} + K_x R e^{-ik_x \Delta x/2}) = -\hat{K}'_x T$$

We observe that, in the continuum limit ($\Delta x \rightarrow 0$), this relationship becomes equivalent to the standard one.

It is then trivial to extract the reflection coefficient according to (12). The dispersion relation for elastic and lossy medium provides an additional relationship between horizontal wavenumbers k_x and k'_x

$$\sin(\delta') = (1 - iR_x) \sin(\delta)$$

with $\delta = k_x \Delta x/2$ and $\delta' = k'_x \Delta x/2$, whereas vertical wavenumbers are the same in the two media for phase matching (Chew and Liu, 1996). Upon substitution we obtain

$$R = \frac{e^{-2i\delta} - 1 - e^{i\delta'} (e^{i\delta} - e^{-i\delta})}{1 - e^{2i\delta} - e^{i\delta'} (e^{i\delta} - e^{-i\delta})}. \quad (16)$$

Consistency of the solution can be verified by removing the lossy property of the PML ($\delta' = \delta$) and verifying that in this case $R = 0$.

R can be considered as a function of the variable δ only, or, alternatively, of $\pi/\delta = \lambda/[\Delta x \cos(\theta)]$, that is, the number of points per wavelength (ppw) versus the cosine of the incidence angle. Figure 1a represents the variation of R as a function of $\lambda/\Delta x$ (ppw) at perpendicular incidence. At different incidence angles, function (16) and the relative graphic can be interpreted as a function of the wavelength λ stretched along the x axis (according to the appropriate projection on the wavenumber along the normal to the boundary), reaching maximum deformation for vertical incidence, for which the cosine becomes unit. First, for $\Delta x \rightarrow 0$ the reflection coefficient becomes effectively zero, so that the continuum limit is met. This is an important statement, for it is a formal demonstration that the discrete PML formulation does converge to the appropriate continuum behavior. If one supposes that Δx and the incidence angle are fixed, then the curve represents the reflection coefficient as a function of the wavelength only. The curve decreases rapidly up to 10 ppw, then continues to diminish more slowly at greater wavelengths. In any case, the reflection coefficient remains about a hundred times smaller than the incident signal. For

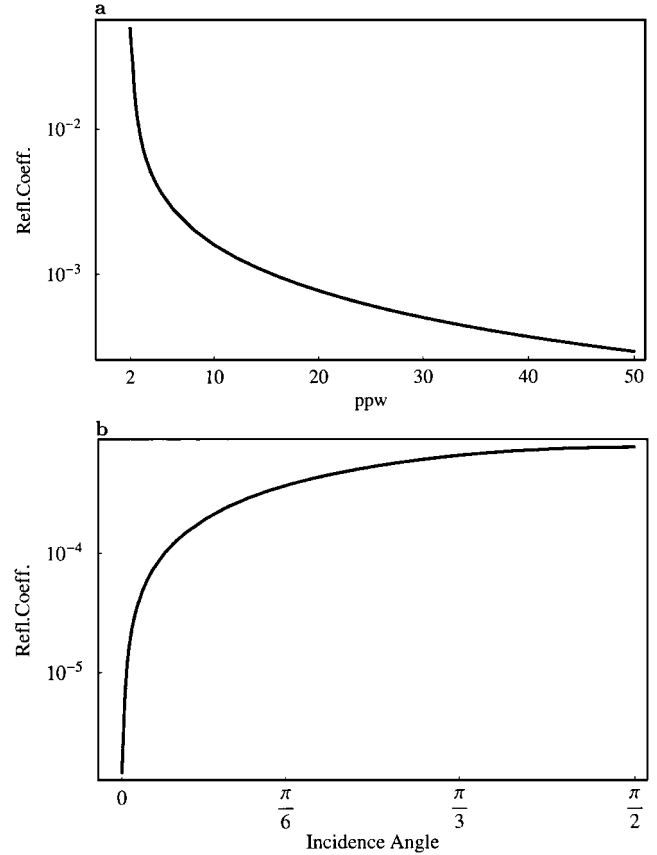


Figure 1. SH -reflection coefficient at a PML interface, computed analytically according to equation (16) on a discrete finite-difference grid (see details in text). (a) Reflection for perpendicularly incident wave, as a function of points per wavelength (ppw); (b) reflection coefficient for a number of ppw fixed at 15, as a function of the incidence angle.

clarity, we also represent the variation of R as a function of the incidence angle for points per wavelength fixed at 15 (Fig. 1b). In this analysis, Δx , c_s , and Δt are fixed, yielding a Courant number $P = 0.3$, a classic value used in the standard fourth-order finite-difference scheme to avoid numerical instability.

The present analysis considers a PML with a constant $\alpha(x)$ damping function; this, in fact, corresponds to isolating the contribution of only the first node inside the PML in the case of inhomogeneous $\alpha(x)$. For an accurate evaluation of the reflections cumulated from the whole thickness of the PML with nonconstant $\alpha(x)$ damping, we shall resort to numerical tests in the next sections.

The problem is more complex for a plane P wave, because we can have conversions into SV waves. We preserve the hypothesis and the configuration of the previous analysis, except that here we consider a P wave instead of an SH wave. The equations regarding this case are developed in Appendix B. By solving the linear system described by equations (24–27), in combination with the definition of the attenuating wavenumber $\hat{K}$ (equation 11), under the hypoth-

esis that the variations of the P -transmission angle with respect to the P -incidence angle can be neglected (they represent a second-order effect), the reflection/transmission coefficients are readily obtained. We do not expand the rather lengthy expressions here, but we note that the S -reflected and S -transmitted wave are utterly negligible at any incidence angle. Consequently, the P -reflection curve becomes similar to the SH curve (although it is obtained by imposing slightly different continuity conditions), and we can refer to the SH graph (1) for the trend of the P -reflection coefficients as a function of the wavenumber and the incidence angle. However, the range of allowable k_x^p values is constrained by the existence of k_x^s .

The same results apply to the case of an SV -incident wave, that is, the P -reflection and transmission coefficients are both negligible, and Figure 1 can also be interpreted as the SV -reflection curve.

In conclusion of this section, we note that PML efficiency increases with increasing wavelength. We recall that the preceding analysis is done in the framework of a discrete PML of constant damping and infinite thickness; the role of inhomogeneous damping and finite-layer thickness will be accounted for in the numerical study of the next section.

Numerical Tests for Body Waves

The efficiency of PML depends mainly on three factors: the PML interface is realized in a discrete medium, where the fields are not defined in each point of the space, the finite thickness of the layer, and the numerical dispersion produced in the same PML. The first effect was evaluated analytically in the previous section, at least in the case in which the function $\alpha(x)$ is constant in the layer. The effect of layer thickness has also been evaluated analytically by (Collino and Tsogka, 2001), who demonstrated that the attenuation of the signal in the PML is

$$\frac{\|\mathbf{U}_{\text{out}}\|}{\|\mathbf{U}_{\text{in}}\|} \propto e^{-2 \cos \theta / c \int_0^1 \alpha(s) ds}$$

where $\mathbf{U}_{\text{in}}$ and $\mathbf{U}_{\text{out}}$ are the incoming and the outgoing fields from PML, respectively, θ is the incidence angle, c is the velocity of the phase (P or S) that needs to be attenuated by the PML, and the $\alpha(x)$ function is normalized so that the thickness of the PML corresponds to $s = 1$.

In this relationship no frequency dependence appears and the attenuation depends just on the path of the wave, so that we expect a minimum attenuation at normal incidence, for which the route of a wave traveling back and forth through the PML is minimal.

Finally, the dispersion relationship has also been evaluated analytically stability (see Analysis), but the complex relationship between the frequency and the wavenumber doesn't allow the equation to be solved in closed form. On the other hand, because we use a second-order approxima-

tion inside the PML instead of a fourth-order, some of the wavelengths that are close to the minimum sampling inside the model are not well resolved inside the PML, and prominent disturbances may be reflected toward the medium when we consider a signal close to the limit frequency. Such an effect, difficult to quantify analytically, will nonetheless be implicitly accounted for in the next section.

To study the numerical effect of the PML on a signal, we compare a signal recorded inside the medium with one obtained on the PML interface, for 2D P -SV simulations. This allows us to use very large grids ($n_x = n_y = 1000$) and thus to separate the effects of a single perfectly matched layer from those, arriving at later times, that may originate at other model edges.

In this example, the parameters have the following dimensional values: $\rho = 2.4 \text{ g/cm}^3$, $\lambda = \mu = 12.81 \cdot 10^6 \text{ dyne cm}^2$ (which corresponds to $c_P = 4 \text{ km/sec}$ in a Poisson medium). Space and time steps are $\Delta x = 0.92 \cdot 10^{-2} \text{ km}$ and $\Delta t = 1.0 \cdot 10^{-3} \text{ sec}$, respectively, so that frequencies up to 20 Hz are well resolved in the standard fourth-order finite-difference scheme. The distance between source and receiver is 300 grid points, the thickness of the PML is 10 grid points. The imposed source is a Gaussian function in time,

$$\frac{1}{\tau_b} e^{-(t-3\tau_b)^2/\tau_b^2} \quad (17)$$

with a flat spectrum up to the corner frequency at about $1/\tau_b$. We consider either an explosive source (obtained by soliciting the diagonal stresses at one node) or a pure shear source (obtained by soliciting the four velocity nodes around a shear stress node).

Obviously, the appropriate choice of a $\alpha(x)$ damping function is a key point for the efficiency of the PML. Two main parameters are considered: the constant A and the exponent n in the power law (equation 8). Analytically, in a continuous medium, the greater is the attenuation, the greater is A and, on the contrary, the smaller is n . In fact, the exponential is proportional to the integral of $\alpha(x)$, which decreases with n , if the integration interval is $[0,1]$. On the other hand, in a discrete medium we must resort to a compromise, because increasing A (or decreasing n) also increases the contrast between a grid node and the next one, eventually producing significant spurious reflections.

In our study we have evaluated the Fourier spectrum of the difference between the signal sampled along the PML-medium interface and the one in the medium. The amplitude of the signal is normalized for the plateau value of the source spectrum and the geometrical spreading. The final spectrum represents the cumulative reflection effects from PML and is represented as the reflection coefficient versus points per wavelength.

The graph of Figure 2 shows that the choice of the PML parameters is critical. In particular, for a value of $A = 2.5$ and $n = 5$, the reflection of the P waves is independent by

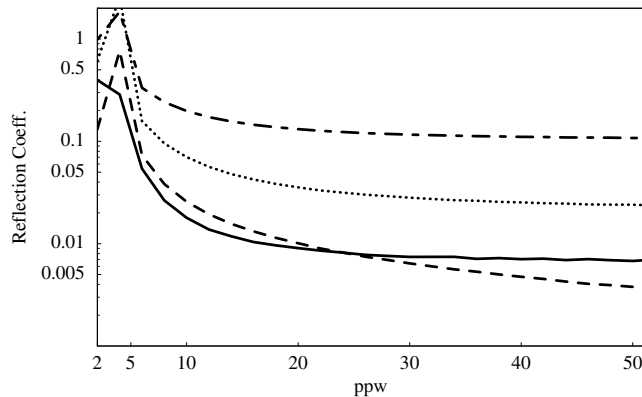


Figure 2. Numerically estimated reflection from PML vs points per wavelength for a *P*-wave incidence. The various curves correspond to different coefficients affecting the lateral variation in the dissipation function $\alpha(x)$: $n = 5$, $A = 2.5$ (solid curve); $n = 5$, $A = 4.5$ (dashed curve); $n = 3$, $A = 2.5$ (dotted curve); $n = 2$, $A = 2.5$ (dash-dot curve). The first two cases are consistently better than the others.

the frequency for more than 10 ppw, and the reflection from PML is about six per thousand with respect to the incoming signal. If a greater value of A is chosen, a less regular curve is observable (dashed line in Fig. 2). In this case we have a better attenuation for $\text{ppw} > 20$, whereas for $\text{ppw} < 10$ the function increases remarkably, until reaching a peak of 0.7 for 5 ppw. On the hard, the sensibility of the PML to the the exponent n of the power law of $\alpha(x)$ is greater. If we choose $n = 3$, for the same value of A , the curve has a plateau level five to six times greater than for $n = 5$. Finally, we show the reflection obtained with the same exponent as used in Collino and Tsogka (2001) ($n = 2$). In the latter case, the reflection is stronger and could be reduced only by increasing the number of nodes inside the PML.

The *S* case is more critical; the curve goes below 1% starting from 15 ppw and drops down to 0.2% for 50 ppw (Fig. 3).

Finally, in Figure 4 we analyze the trend of the reflection coefficient for a fixed range of frequencies ($f < 5$ Hz) with the incidence angle. As expected it increases, as the incidence angle increases, in relation to the effective way done by the wave in the layer. The asymptotic value is of about 1.1% for vertical incidence.

Numerical Tests for Rayleigh Waves

Because PML is, in fact, a nonstandard sponge, there is in principle no limit to the type of wave it can be applied to; indeed, we find that it attenuates surface waves efficiently, as well. This constitutes the major advantage of the PML with respect to standard paraxial methods. In this section we evaluate the efficiency of the PML with Rayleigh waves, and compare it with the paraxial method proposed by Higdon (1991). For this study we consider a homogeneous medium

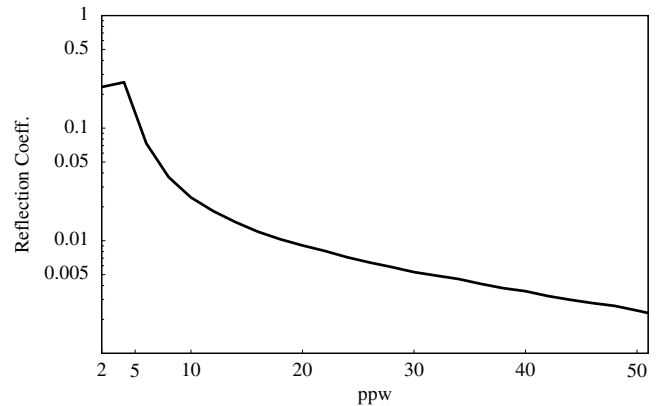


Figure 3. Numerically estimated reflection from the PML vs points per wavelength for a perpendicular *S* wave. In this example, the coefficients of the dissipation function $\alpha(x)$ were fixed at $n = 5$ and $A = 2.5$.

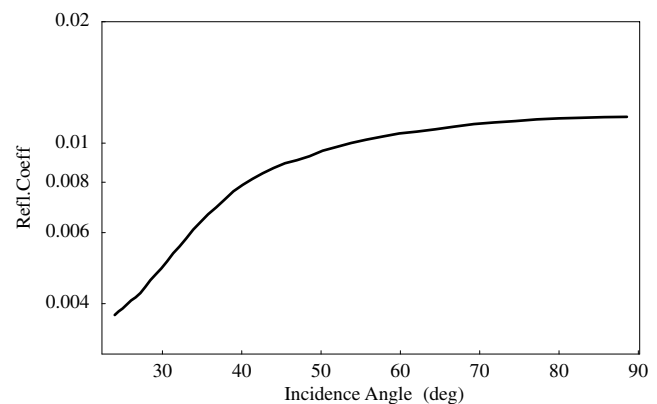


Figure 4. Maximum PML reflection coefficient relative to the frequency as a function of the incidence angle, for an incident *P* wave. In this example, the coefficients of the dissipation function $\alpha(x)$ were fixed at $A = 2.5$ and $n = 5$.

bounded by a free surface, with the same properties as in the previous section. Space and time steps are also unchanged. The z axis is orthogonal to the surface and the model dimension are $n_x = 1084$ and $n_y = 285$. The PML parameters are $A = 2.5$, $n = 5$, with a thickness of 10 grid points. An explosive source with duration $\tau_b = 0.4$ sec is located 10 grid points below the surface, a depth negligible with respect to the source–receiver distance to maximize the amplitude of Rayleigh waves. For a fixed source–receiver distance, two configurations are considered: (1) the receiver is in the middle of the medium, so that no reflections from the boundaries can reach it within the duration of the computation; (2) the receiver is located right along the medium–PML interface, whereas the source is translated to preserve its distance from the receiver. Thus, the difference between signal 1 and signal 2 is a measure of the PML effect only. The Rayleigh wave train is selected and isolated from other arrivals (direct *P* or converted *S*) for the analysis. These operations were repeated

for a medium with paraxial absorbing conditions (Higdon, 1991), so that both methods could be compared.

In Figure 5 the reflection effects are plotted as function of the number of points per wavelength. We observe that the PML absorbing boundaries are more efficient with respect to the paraxial method for the Rayleigh waves. For $\text{ppw} > 10$ the PML reflection becomes smaller than 2% and for $\text{ppw} > 20$ it is below 1%. On the contrary, if the minimum for the paraxial method (Higdon, 1991) is obtained at 8 ppw and corresponds to 12% of the incoming signal, then for $\text{ppw} > 10$ the reflection coefficient increases up to 30% for 50 ppw.

3D Examples

Homogeneous Half-Space

We tested the efficiency of PML for the 3D model with free-surface boundary conditions. We adopted homogeneous media, in which analytical solutions are known, to isolate the effects of PML. For the test, we compare a model in which the receiver is located right against the PML to another model in which the receiver is far from the PML. The same source-receiver distance is maintained in both models, and the receivers are two grid nodes below the free surface. The elastic parameters are normalized ($\lambda = 1 = \mu = \rho = 1$), and the time and space grid step are consequently scaled to respect the stability and dispersion conditions ($\Delta x = 0.2$, $\Delta t = 0.05$ in dimensionless units). The model is a regular box with dimensions ($n_x = 128$, $n_y = 128$, $n_z = 64$). An explosive point source near the surface is located at node (64, 32, 8), the source function is a Gaussian (17) with characteristic time $\tau_b = 0.7$ sec. The parameters of the decay function $\alpha(x)$ (equation 8) are $A = 4.5$ and $n = 2$. The increasing of n increases the amplitude of the reflection wave.

The seismograms in Figure 6 are related to the first arrival: the upper trace represents a record for a station far from the PML, whereas the middle trace corresponds to a receiver on the elastic-lossy interface. The lower trace is the difference between the two signals (all the traces are normalized at the maximum of the first signal). The maximum value of the difference is 0.5%, which ensures the efficiency of the PML in 3D media, at least for body waves.

The efficiency of analytical PML conditions increases with incidence angle. We show a sequence of snapshots (Fig. 7), which represent the modulus of the velocity in a y - z plane, traced in correspondence of the source (at $n_x = 64$). All the plots are normalized at the same scale; we can observe a complete absorption of the body and the surface waves, at a variety of incidence angles; to be visible, reflections from the PML should be magnified 100 times.

Half-Space with an Interface

Having tested the PML method for Rayleigh waves in a 3D medium, we add a horizontal interface with a transition to stiffer parameters to create an internal contrast. Our aim is to test the efficiency of PML in the presence of inhomogeneities other than the free surface and, in this case, whether PML works also with the evanescent waves that propagate along the interface.

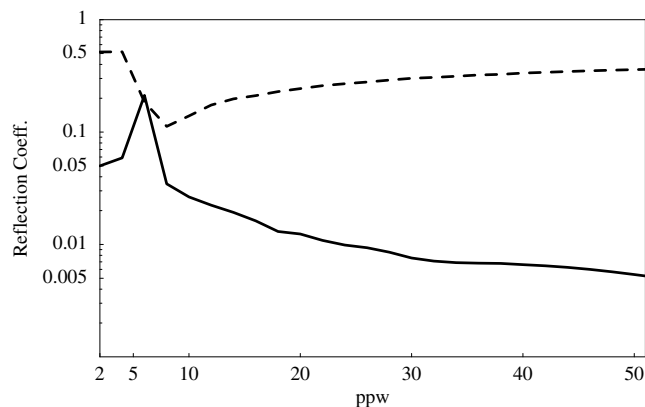


Figure 5. Comparison between the PML absorbing boundary conditions (solid line) and the Higdon's (1991) paraxial conditions (dashed line), in the case of a Rayleigh wave. The reflection from one border is plotted as function of the number of points per wavelength. PML parameters are $n = 5$, $A = 2.5$ and the layer thickness is 10 nodes.

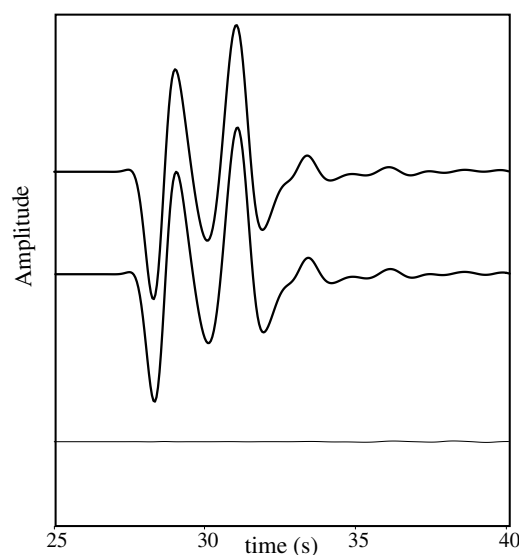


Figure 6. Comparison between unbounded medium and PML interfaced medium for longitudinal waves. The top trace is obtained far from the border, whereas the middle trace is obtained right at the interface with the PML. The bottom trace represents the difference in the same vertical scale, that is, it is an estimate of the spurious reflection from the PML. (In this example reflection is of the order of 0.5% of the incident signal.) See text for details.

geneties other than the free surface and, in this case, whether PML works also with the evanescent waves that propagate along the interface.

The test is conducted for a 3D medium with properties ($\lambda = 1 = \mu = \rho = 1$) down to a depth of 32 grid nodes, below which the medium properties are set to ($\lambda = \mu = 1.7$; $\rho = 1$). In this case we vertically align seven receivers,

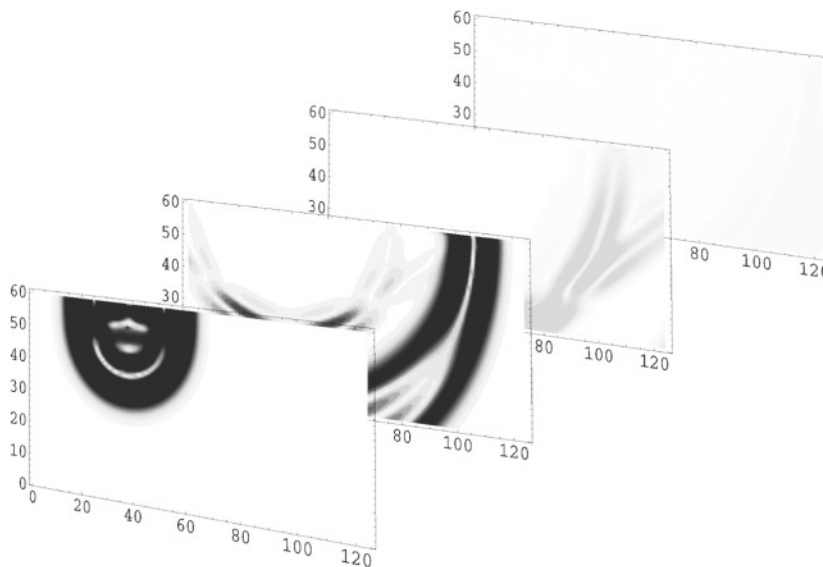


Figure 7. Snapshots of 3D propagation with P , S , and Rayleigh waves reaching the PML boundary.

as shown in Figure 8 across the interface so that the fourth receiver falls on the horizontal interface itself. As in the preceding section, we start by locating the receiver column against the PML boundary; then we moved away the PML so that the receiver column remained far from all the absorbing boundaries, but stayed at the same distance from the source. We then had two sets of seismic traces, which would be identical if the PML condition was indeed perfect. The two sets of traces are plotted, superimposed, in Figure 9. The match is almost within the thickness of the curve (the maximum amplitude mismatch is of the order of 0.7%).

Finally, we show a sequence of four snapshots representing the modulus of the velocity in a y - z plane, traced in correspondence of the source (at $n_x = 64$), with a common scale (Fig. 10). Both Rayleigh and evanescent (interface) waves are seen to vanish at the medium boundary.

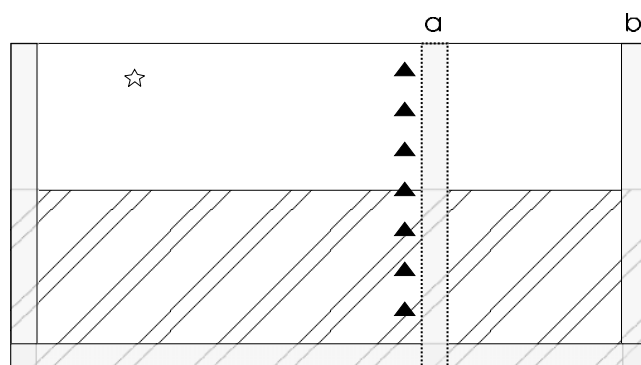


Figure 8. Model indicating relative position of receivers, interface, and PML for the 3D simulations with both free surface and interface. (a) Simulation with the PML in the immediate vicinity of the receivers; (b) simulation with the PML moved away from the receivers.

Conclusions

We showed that the spurious reflections introduced because of the discretization of the PML–medium interface could be estimated analytically by an extension of the classical technique for computing the reflection–transmission coefficients at an interface. With such analysis, we find that as the space sampling tends to zero, the spurious reflections vanish in the numerical scheme. As a result, we can formally confirm the equivalent behavior of the discretized and continuous PML formulations in the limit of the continuum.

Moreover, a comprehensive numerical evaluation of all

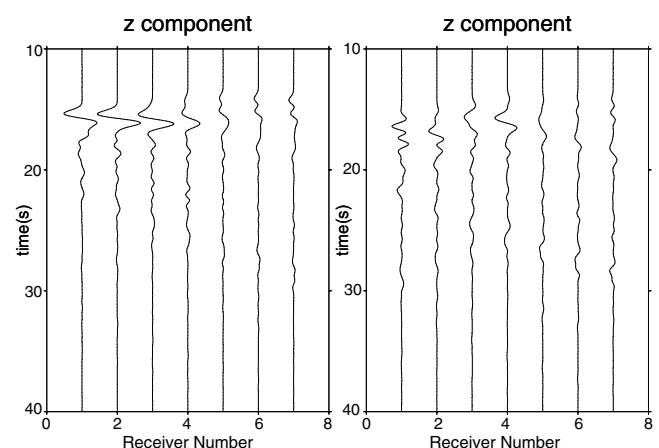


Figure 9. Velocigrams from simulations of the 3D half-space with contrasting interface (cf., Fig. 8). The traces obtained with receivers close to the PML (solid curves) and far from the PML (dotted curves) are superimposed for evaluation of the spurious reflection. The maximum difference is of the order of the curve thickness (almost invisible in this graphic), that is, about 0.7% of the signal amplitude.

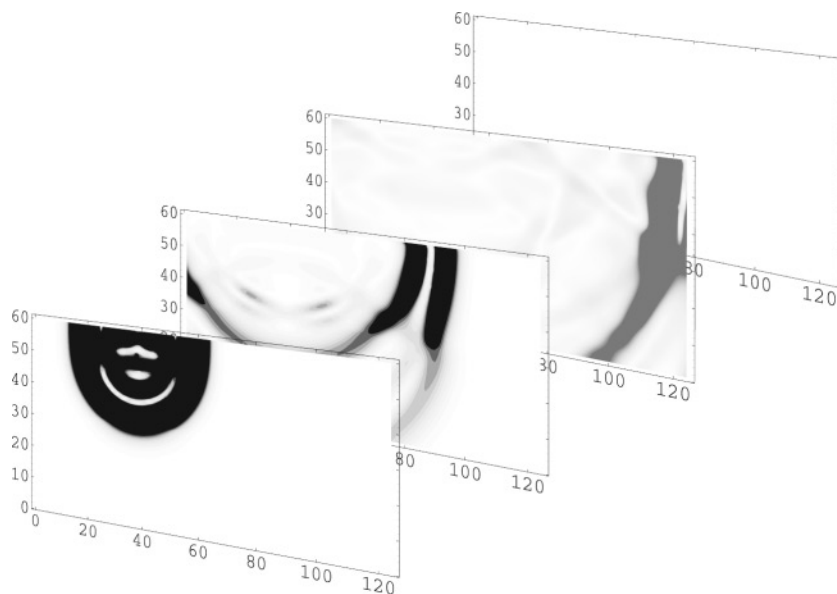


Figure 10. Snapshots of the 3D simulations with both free surface and interface. (The gray scale indicates the modulus of particle velocity.) In the third snapshot the conical wave propagating upward from the interface is seen as a straight wavefront.

spurious reflections shows that: (1) Reflections increase with frequency owing to numerical dispersion, but they remain negligible in the whole useful range of points per wavelength of about 8 ppw and above. (2) There is no long wavelength limit to the efficiency, indeed the reflection remains stable and negligible even at wavelengths far above the thickness of the PML layer. This point is surprising and very important, because it has several practical consequences for numerical modeling; for example, there is no need to scale the thickness of the PML layer with the size of the model. (3) As far as angular dependence is concerned, the method performs well for perpendicular incidence (the worst cases yield reflections of about 1%) and even better as the incidence becomes shallower; it is very effective even on inhomogeneous waves such as (4) Rayleigh waves at the free surface of a half-space and (5) inside inhomogeneous media even in the presence of strong contrasts where interface waves develop.

In conclusion, the PML method has several advantages over previous methods that have been used extensively to date (such as the paraxial approximation), namely, it remains effective possibly under any practical circumstance (e.g., shallow or acute angles of incidence, homogeneous or inhomogeneous waves, long or short wavelengths, heterogeneous media). Besides, the additional computational cost introduced by the extra layers and float-point operations is negligible and remains such even for large models.

Acknowledgments

We thank C.-R. Calidonna and M. M. Furnari from Istituto di Cibernetica, C.N.R., Naples (Italy) for guidance in program optimization and for granting extensive access to the enhanced computing facilities of their

Institute. We are grateful to A. Zollo from Scienze Fisiche, University of Napoli, for hosting and financial support.

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Appendix A

The stability relationship

$$\sin^2(\omega \Delta t/2) = \frac{c^2 \Delta t^2}{\Delta x^2} \left(\frac{\sin^2(k_x \Delta x/2)}{(1 - iR_x)^2} + \sin^2(k_y \Delta x/2) + \sin^2(k_z \Delta x/2) \right) \quad (18)$$

relates the pulsation ω to the wavenumber $\mathbf{k}$. In standard finite differences, the relaxation factor R_x disappears and the upper bound for the stability coefficient P can be written as:

$$P \leq 1 \quad \text{where } P = \frac{c \Delta t}{\Delta x \sqrt{3}},$$

which guarantees the stability of the method for any couple $(\omega, \mathbf{k})$ verifying equation (18).

With the PML formulation, the problem seems *a priori* more complex because it is not straightforward enough to obtain a real solution for both ω and $\mathbf{k}$ from equation (18). Letting the PML boundary lie perpendicular to the x axis, let us first proceed to search for real solutions in ω , k_y , and k_z . Because the first member of equation (18) is real, it is required that the second member be real also. Supposing that

$$k_x \Delta x/2 = x + iy$$

and recalling the hyperbolic-trigonometric relationship

$$\sin(k_x \Delta x/2) = \sin(x) \cosh(y) + i \cos(x) \sinh(y) \quad (19)$$

we can equate the imaginary part of equation (18) to zero and obtain:

$$\begin{aligned} [\cos(x) \sinh(y) + R_x \sin(x) \cosh(y)] \\ [-R_x \cos(x) \sinh(y) + \sin(x) \cosh(y)] = 0 \end{aligned} \quad (20)$$

where $R_x = \alpha \Delta t/2 \cot(\omega \Delta t/2)$. Equation (20) has two solutions in R_x ,

$$-\frac{\tanh(y)}{\tan(x)} = R_x \quad (21)$$

$$\frac{\tanh(y)}{\tan(x)} = \frac{1}{R_x}. \quad (22)$$

To avoid the plane wave increasing indefinitely, the solutions for k_x in the positive complex half-plane must be discarded. Because we consider a progressive wave in the positive x direction, only solution (21) remains.

Furthermore, equation (19) can be written as:

$$\sin(k_x \Delta x/2) = \sin(x) \cosh(y) + i \cos(x) \cosh(y) \tanh(y)$$

where $\tanh(y)$ can be substituted according to (21). After introducing P and expanding the expression for R_x , we obtain:

$$\sin^2(\omega \Delta t/2) = \frac{1}{3} P^2 \left(\frac{\sin^2(x)}{1 - \frac{1}{3} \rho^2 P^2 \cot^2(\omega \Delta t) \tan^2(x)} + \sin^2(k_y \Delta x/2) + \sin^2(k_z \Delta x/2) \right),$$

where ρ is grouping several terms independent of P . Using the upper bound $\sin(k_y \Delta x/2) = \sin(k_z \Delta x/2) = 1$ for the two end terms and posing $w = \sin^2(\omega \Delta t/2)$, we get waves as well as body waves

$$w - \frac{P^2 \sin^2(x) w}{3w - (1 - w) \rho^2 P^2 \tan^2(x)} - \frac{2}{3} P^2 = 0. \quad (23)$$

Because w is the square of a Sine function, equation (23) is consistent provided that $w \leq 1$. Thus, we can search for a threshold value of P below which the condition $w \leq 1$ is verified for any x . We first note that, for $x = 0$, a solution is

$$w = \frac{2}{3} P^2.$$

Moreover, in the limit $x \rightarrow \pi/2$ we have two solutions:

$$w = \frac{2}{3} P^2; \quad w = 1.$$

For $0 < x < \pi/2$, equation (23) has two real, continuous solutions in x . Existence of a solution in $x = 0$ imposes that $\frac{2}{3} P^2 < 1$. Because the functions are continuous, discarding any solution $w = 1$ also forbids that $w > 1$ in the interval $0 < x < \pi/2$. Thus, we want to exclude solutions obtained by setting $w = 1$ in equation (23), of the form

$$\sin(x) = \frac{3 - 2P^2}{P^2}$$

This equation has no solution for $x < \pi/2$ provided that:

$$\frac{3 - 2P^2}{P^2} \geq 1,$$

Finally, we obtain

$$P^2 \leq 1,$$

that is, the same stability condition as in standard finite differences.

Appendix B

The velocity field in the elastic medium can be expressed as:

$$\mathbf{v} = \mathbf{d} e^{i(\omega t + k_x^p x - k_z^p z)} + R_{PP} \mathbf{d}^p e^{i(\omega t + k_x^p x - k_z^p z)} + R_{PS} \mathbf{d}^{rs} e^{i(\omega t + k_x^s x - k_z^s z)},$$

where $\mathbf{d}$ represents the director cosine of the incident wave, $\mathbf{d}^p$ that of the reflected P , and $\mathbf{d}^{rs}$ that of the reflected S . R_{PP} and R_{PS} are the P and S reflection coefficients. Analogously, the expression for the velocity field inside the PML is

$$\mathbf{v}' = T_{PP} \mathbf{d}^p e^{i(\omega t - k_x^p x - k_z^p z)} + T_{PS} \mathbf{d}^{ts} e^{i(\omega t - k_x^s x - k_z^s z)}$$

where T_{PP} and T_{PS} express the P and S transmission coefficients, whereas $\mathbf{d}^p$ and $\mathbf{d}^{ts}$ are the director cosines of the transmitted waves. The superscripts p or s identify the phase to which the wavenumber $\mathbf{k}$ is referred. We need four conditions to obtain the reflection/transmission coefficients. With our choice of reference frame, only σ_{xx} and v_z are located at the interface, so the continuity of these two quantities can be imposed. Two more equations can be obtained by imposing the conditions on the derivatives of v_x and σ_{xz} , which have to satisfy momentum and constitutive laws, respectively, as in the SH case.

In all the equations the director cosines and the wavenumbers are supposed positive. Continuity of v_z yields to the equation:

$$d_z + R_{PP} d_z - R_{PS} d_z^{rs} = T_{PP} d_z^p - T_{PS} d_z^{ts} \quad (24)$$

Whereas continuity of σ_{xx} implies:

$$\begin{aligned} & (\lambda + 2\mu) (K_x^p d_x + K_x^p R_{PP} d_x + K_x^s R_{PS} d_x^{rs}) \\ & + \lambda (K_z^p d_z + K_z^p R_{PP} d_z + K_z^s R_{PS} d_z^{rs}) \\ & = (\lambda + 2\mu) (\hat{K}_x^p T_{PP} d_x^p + \hat{K}_x^s T_{PS} d_x^{ts}) \\ & + \lambda (K_z^p T_{PP} d_z^p + K_z^s T_{PS} d_z^{ts}). \end{aligned} \quad (25)$$

As mentioned previously, two additional conditions on the derivatives of v_z and σ_{zz} can be obtained through the derivative of the momentum law and the constitutive equation,

$$\partial_{xt}^2 \sigma_{xz} = \frac{\mu}{\lambda + 2\mu} \partial_{xt}^2 \sigma_{xx} + \frac{\lambda\mu}{\lambda + 2\mu} \partial_{yy}^2 v_y + \mu \partial_{xx}^2 v_y$$

$$\partial_x v_x = \frac{1}{\lambda + 2\mu} \partial_t \sigma_{xx} - \frac{\lambda}{\lambda + 2\mu} \partial_z v_z$$

where the x derivatives at the left-hand member are evaluated as the finite difference between two nodes straddling the interface, whereas the right-hand member is defined on the interface.

The discrete version leads to:

$$\begin{aligned} & iK_x^p (d_x K_z^p + R_{PP} d_x K_z^p + d_z K_x^p + R_{PP} d_z K_x^p) \\ & + iK_x^s (R_{PS} d_x^{rs} K_z^s - R_{PS} d_z^{rs} K_x^s) \\ & = \frac{1}{\Delta x} (d_x K_z^p e^{ik_x^p \Delta x/2} - R_{PP} d_x K_z^p e^{-ik_x^p \Delta x/2} \\ & - R_{PS} d_x^{rs} K_z^s e^{-ik_x^s \Delta x/2} + d_z K_x^p e^{ik_x^p \Delta x/2} \\ & - R_{PP} d_z K_x^p e^{-ik_x^p \Delta x/2} + R_{PS} d_z^{rs} K_x^s e^{-ik_x^s \Delta x/2} \\ & - T_{PP} d_z^p \hat{K}_x^p e^{-ik_x^p \Delta x/2} - T_{PS} d_z^{ts} \hat{K}_x^s e^{-ik_x^s \Delta x/2} \\ & - T_{PP} d_x^p K_z^p e^{-ik_x^p \Delta x/2} + T_{PS} d_x^{ts} K_z^s e^{-ik_x^s \Delta x/2}) \end{aligned} \quad (26)$$

and

$$\begin{aligned} & iK_x^p d_x (1 + R_{PP}) + K_x^s d_x R_{PS} \\ & = \frac{1}{\Delta x} (d_x e^{ik_x^p \Delta x/2} - R_{PP} d_x e^{-ik_x^p \Delta x/2} - R_{PS} d_x^{rs} e^{-ik_x^s \Delta x/2} \\ & - T_{PP} d_x^p e^{-ik_x^p \Delta x/2} + T_{PS} d_x^{ts} e^{-ik_x^s \Delta x/2}) \end{aligned} \quad (27)$$

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Manuscript received 2 April 2002.